

The Economic Problem

Based on Ch. 2, Economics 10th ed., Parkin, Powell & Mathews

Course: Eco 101 (Introduction to Microeconomics)

Production Possibilities Frontier (PPF)

Definition of the model:

The PPF represents the **possible combinations** of two goods that can be produced in a **certain period of time** under the conditions of a **given state of technology** and **fully employed resources**.

Assumption:

- The economy produces only two goods
- Full employment
- Fixed resources and technology

Continued

The **production possibilities frontier (PPF)** is the boundary between those combinations of goods and services that can be produced and those that cannot. To illustrate the PPF, we focus on two goods at a time and hold the quantities produced of all the other goods and services constant. That is, we look at a model economy in which everything remains the same (*ceteris paribus*) except for the production of the two goods we are considering.

The production possibilities frontier for Cola and pizza shows the limits to the production of these two goods, given the total resources available to produce them.

The model is presented in the next slide.

Production Possibilities Frontier

Possibility	Pizzas (millions)		Cola (millions of cans)
A	0	and	15
B	1	and	14
C	2	and	12
D	3	and	9
E	4	and	5
F	5	and	0

The table lists six production possibilities for cola and pizzas. Row A tells us that if we produce no pizzas, the maximum quantity of cola we can produce is 15 million cans. Points A, B, C, D, E, and F in the figure represent the rows of the table. The curve passing through these points is the **production possibilities frontier (PPF)**.

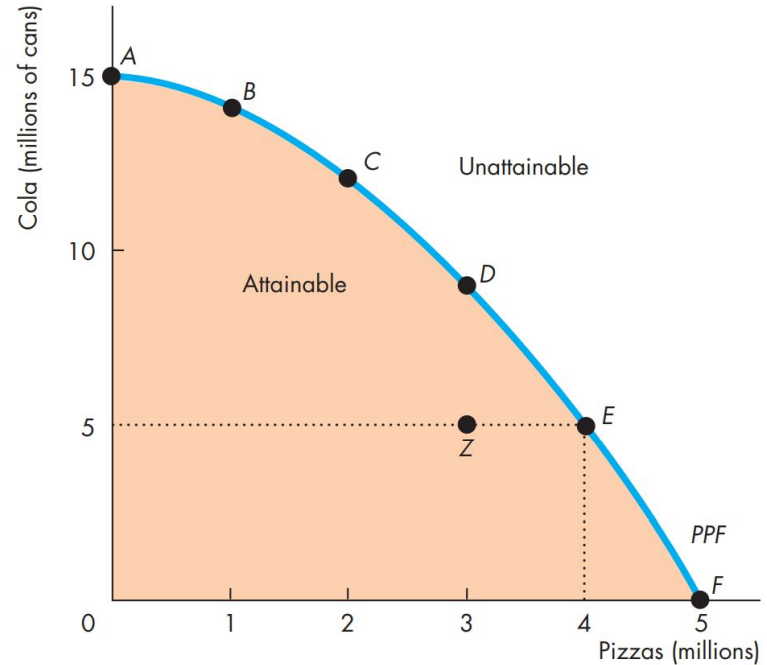


Figure 2.1 shows the PPF for two goods: cola and pizzas.

Continued

- The PPF illustrates **scarcity** because we cannot attain the points outside the frontier. These points describe wants that can't be satisfied. These points are unattainable.
- Production is possible at any point inside the orange area or on the frontier. These points are attainable. Points inside the frontier, such as point Z, are **inefficient** because resources are *wasted* or *misallocated*. At such points, it is possible to use the available resources to produce more of either or both goods.
- We achieve **production efficiency** if we cannot produce more of one good without producing less of some other good. Points on the frontier are efficient.

Trade-off Along the PPF

Every choice along the PPF involves a trade-off – we must give up something to get something else. On the PPF in the Figure, we must give up some Cola to get more Pizza or give up some Pizza to get more Cola.

At any given point in time, we have a fixed amount of labor, land, capital, and entrepreneurship. By using our available technologies, we can employ these resources to produce goods and services. But we are limited in what we can produce. This limit defines a boundary between what we can attain and what we cannot attain.



Opportunity Cost

The PPF helps us to make the concept of opportunity cost precise and enables us to calculate it.

The opportunity cost of producing an additional pizza is the number of Cola we must forgo. Similarly, the opportunity cost of producing an additional Cola is the quantity of pizzas we must forgo. For example, at point C in the Figure, we produce fewer pizzas and more Cola than at point D.

If we choose point D over point C, the additional 1 million pizzas cost 3 million Cola. One pizza costs 3 Cola. So 1 Cola costs $\frac{1}{3}$ of a pizza.

Law of Increasing Opportunity Cost

Increasing Opportunity Cost: In the real-world people have **varying (different) abilities or skills**.

For example, some labor are good at making Cola and some are good at making Pizza. When we produce a small number of Pizza, we will use the labor who are good at making Pizza. However, as we produce more and more Pizza we have to hire the labor who are good at making Cola and so we will have to sacrifice more Cola to produce Pizza. Hence, the opportunity cost of producing Pizza increases as we produce more Pizza.



Shifting of PPF: Economic Growth

Economic Growth

- **Definition:**
Economic growth = increase in the productive capacity of an economy.
- **Graphical Representation:**
Shown by an **outward shift of the Production Possibilities Frontier (PPF)**.

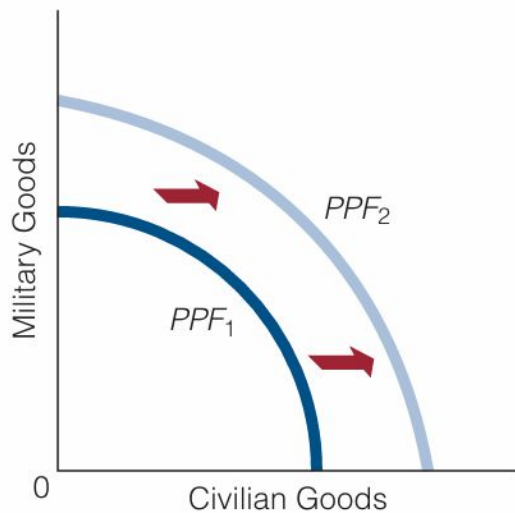
Key Factors Driving Economic Growth

1. **Increase in Quantity of Resources**
 - More resources (e.g., new discoveries) → higher production potential.
 - Results in outward shift of PPF (e.g., from PPF_1 to PPF_2).
 - Allows for increased production of **both** military and civilian goods.
2. **Advancement in Technology**
 - Better skills & knowledge in production.
 - Allows:
 - More output with same resources, or
 - Same output with fewer resources.
 - Leads to outward PPF shift, similar to resource increase.

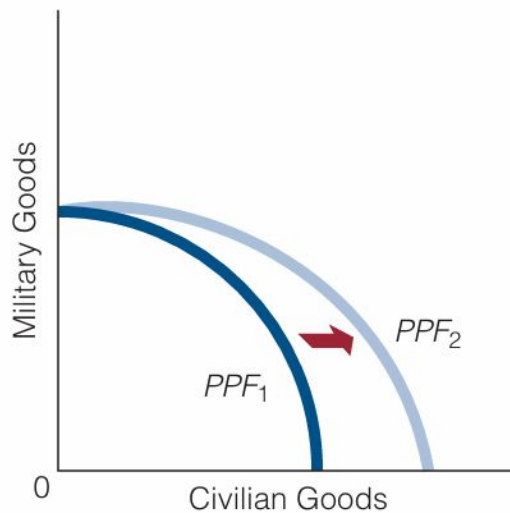
Economic Growth Within a PPF Framework

An increase in resources or an advance in technology (that can lead to more of both goods being produced) can increase the production capabilities of an economy, leading to economic growth and a shift

outward in the PPF, as shown in part (a). If the advance in technology leads to the greater production of only one good (such as civilian goods in this exhibit), then the PPF shifts outward, as shown in part (b).



(a)



(b)

Comparative Advantage and Absolute Advantage

- Comparative Advantage:** A person has a comparative advantage in an activity if they can perform it at a lower **opportunity cost** than others.

- Absolute Advantage:** A person has an absolute advantage if they are **more productive per hour** than others in an activity.

- Key Difference:** Absolute advantage compares **productivity**, while comparative advantage compares **opportunity costs**.

- A person with an **absolute advantage** in multiple activities **may not** have a **comparative advantage** in all of them.

Let's explore the idea of comparative advantage by looking at two smoothie bars: one operated by **Liz** and the other operated by **Joe**.

TABLE 2.1 Liz's Production Possibilities

Item	Minutes to produce 1	Quantity per hour
Smoothies	2	30
Salads	2	30

TABLE 2.2 Joe's Production Possibilities

Item	Minutes to produce 1	Quantity per hour
Smoothies	10	6
Salads	2	30

- Liz's opportunity cost of producing 1 smoothie is 1 salad, and
- Liz's opportunity cost of producing 1 salad is 1 smoothie.

- Joe's opportunity cost of producing 1 smoothie is 5 salads, and
- Joe's opportunity cost of producing 1 salad is 1/5 of a smoothie.

Comparative Advantage: Who Produces What?

Liz's Comparative Advantage

- Liz has a comparative advantage in **producing smoothies**.
- **Opportunity cost of 1 smoothie: 1 salad**
- Joe's opportunity cost of 1 smoothie: **5 salads**

Joe's Comparative Advantage

- Joe has a comparative advantage in **producing salads**.
- **Opportunity cost of 1 salad: $1/5$ of a smoothie**
- Liz's opportunity cost of 1 salad: **1 smoothie**

Each person should specialize in the activity where they have the **lower opportunity cost**, leading to efficient trade.

Absolute Advantage

Who has the absolute advantage on Smoothies?

- Liz has absolute advantage in **producing smoothies**.
- Liz can produce 30 but Joe can only produce 6

Who has the absolute advantage on Salads?

- No one has absolute advantage in **producing salads**.
- They both can produce 30 salads.
- They both have the same level of productivity.

1. The table below displays the daily production of vanilla and chocolate ice creams (in liters) among the top ice-cream producing companies in Bangladesh:

Company	Vanilla Ice Cream	Chocolate Ice Cream
Delight	60	30
Frosty	40	90
Snowy	70	50

- Which company has the absolute advantage in producing vanilla ice cream?
- Which company has the comparative advantage in producing vanilla ice cream?
- Which company has the comparative advantage in producing chocolate ice cream?